The OR gate output is

$$C = A \text{ OR } B$$

We use the + sign to denote the OR function. Therefore,

$$C = A + B$$

Read this as "C equals A OR B".

For instance, if both the inputs are 1

$$C = A + B = 1 + 1 = 1$$

The truth table for OR gate is

Input		Output
A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

Table 2.8 Truth Table for OR Gate

2.9.3 NOT Gate

The NOT gate, called a logical inverter, has only one input. It reverses the logical state. In other words the output C is always the complement of the input. The logical symbol of the NOT gate is

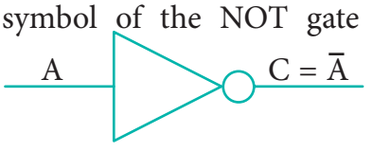


Fig. 2.6 Logic symbol of NOT Gate

The boolean function of NOT gate is

$$C = \text{NOT } A$$

In boolean algerbra, the overbar stands for NOT operation. Therefore,

$$C = \overline{A}$$

Read this as "C equals NOT A" or "C equals the complement of A".

If A is 0,

$$C = \overline{0} = 1$$

On the otherhand, if A is 1,

$$C = \overline{1} = 0$$

The truth table for NOT gate is

Input	Output
A	C
1	0
0	1

Table 2.9 Truth Table for NOT Gate

2.9.4 NOR Gate

The NOR gate circuit is an OR gate followed by an inverter. Its output is "true" if both inputs are "false" Otherwise, the output is "false". In other words, the only way to get '1' as output is to have both inputs '0'. Otherwise the output is 0. The logic circuit of the NOR gateis

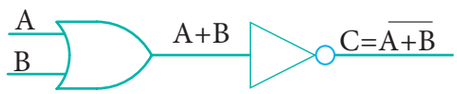


Fig. 2.7 Logic Circuit of NOR Gate

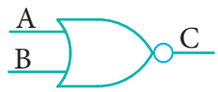


Fig. 2.8 Logic symbol of NOR Gate

The output of NOR gate is

$$C = (\overline{A + B})$$

Read this as "C equals NOT of A OR B" or "C equals the complement of A OR B". For example if both the inputs are 0,

$$C = (\overline{0 + 0}) = \overline{0} = 1$$

The truth table for NOR gate is

Input		Output
A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

Table 2.10 Truth Table for NOR Gate

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